

Bahut badhiya choice! **"Pipeline Paramedic"** ek aisi problem solve karta hai jiska dard har developer samajhta hai. Judges ko impress karne ke liye ye perfect hai. Hamare paas sirf 5 din bache hain, isliye humein straight to the point kaam karna hoga. Chaliye isko ekdam step-by-step todte hain ki aaj aur abhi se kya karna hai.

Phase 1: "The Playground" Setup (Aaj ka kaam)

Sabse pehle humein ek test environment chahiye jahan hum apni fail hoti hui pipeline ko baar-baar test kar sakein.

1. **Ek Naya GitLab Repo Banayein:** GitLab par ek blank project banayein.
2. **Jaan-boojh kar ek Bug likhein:** Koi simple code likhein (jaise Python ya JavaScript mein ek basic calculator) aur usme ek error daal dein.
3. **Pipeline Setup Karein:** Ek `.gitlab-ci.yml` file banayein jo us code ko test kare. Kyunki code mein bug hai, pipeline hamesha **FAIL** honi chahiye.
4. **Tools Install Karein:** Apne system par **GitLab CLI (glab)**, **GitLab Orbit extension**, aur apne pasandeeda code editor (VS Code) ko ready rakhein.

Phase 2: Data Extract Karna (Scripting)

Aapka agent tabhi kaam karega jab uske paas error ki detail hogi. Aapko ek script banani hogi (Python ya Node.js mein) jo ye 3 kaam kare:

1. **Pipeline Status Check:** GitLab API ko call karke check kare ki "Kya koi pipeline fail hui hai?"
2. **Job Logs Fetch Karna:** Agar fail hui hai, toh us specific job ka log (trace) nikaale. Ye log batayega ki error kya tha (e.g., `SyntaxError` ya `AssertionError`).
3. **Code Fetch Karna:** Jis file mein error aaya hai, us file ka current code fetch kare.

Phase 3: GitLab Duo (The Brain) ko Use Karna

Ye aapke project ka sabse main hissa hai. Aapko GitLab Orbit aur Duo Agent Platform ka use karke ek AI prompt bhejna hai.

- **Prompt Design Kuch Aisa Hoga:** "You are an expert developer. Here is the code file: [Insert Code]. The CI/CD pipeline failed with this error log: [Insert Log]. Analyze the log, find the bug in the code, and return ONLY the corrected code without any extra explanation."
- Duo Agent log padhega aur fixed code return karega.

Phase 4: Workflow Automate Karna (The Magic)

Jab Duo aapko theek kiya hua code de de, toh aapke agent ko ye actions lene hain (via API ya CLI):

1. Ek nayi git branch banaye (e.g., `fix/pipeline-error-123`).
2. File ko naye (fixed) code se overwrite karke commit kare.
3. GitLab par us branch ko push kare.
4. Ek **Merge Request (MR)** create kare jiska title ho: *"Auto-Fix: Resolved failing pipeline issue"*.

Phase 5: Orbit Skill mein Convert Karna

Aapki banayi hui script ko ab **GitLab Orbit Skill** ke format mein wrap karna hoga (jaisa unki documentation mein bataya gaya hai) taaki ye Orbit ecosystem ka hissa ban jaye aur easily install ho sake.

Shuruwaat karne ke liye sabse zaroori cheez API aur scripting hai. Aap ye backend logic kis programming language mein likhna prefer karenge—Python ya Node.js (JavaScript)?